

CHAPTER 3

FACTOR ANALYSIS

0.1 Conceptual model Factor Analysis

0.1.1 Introduction

Factor analysis describes, if possible, the covariance relationships among many variables in terms of a few underlying, but unobservable, random quantities called factors.

The general model of FA assumes that the observed variables are linear combinations of a smaller number of latent factors, which are unobserved variables that are assumed to be the true causes of the observed variables.

Additionally, factor model is motivated by the following argument: Suppose variables can be grouped by their correlations. That is, suppose all variables within a particular group are highly correlated among themselves, but have relatively small correlations with variables in a different group. Then it is conceivable that each group of variables represents a single underlying construct, or factor, that is responsible for the observed correlations.

0.1.2 Difference between Principal Components Analysis and Factor Analysis

- i Principal components are linear combinations of the original variables while in FA original variables are expressed as linear combination of the factors.
- ii In PCA, we account for large part of the total variation of the variables while FA we account for contrivances or correlations among variables.

0.1.3 The Orthogonal Factor Model

Let X be an observable random vector, with p components, mean μ and covariance matrix Σ .

The factor model postulates that X is linearly dependent upon a few unobservable random variables $F_1, F_2, F_3, \dots, F_m$, called *common factors*, and p additional sources of variations $\epsilon_1, \epsilon_2, \dots, \epsilon_p$ called *Errors/ residual*.

The general model of a factor analysis is(Johnson & Wichern, 2007):

$$X_1 - \mu_1 = \beta_1 F + \epsilon_1 \quad (1)$$

For a m common factor(s) is;

$$\begin{pmatrix} x_1 - \mu_1 = \beta_{11}F_1 + \dots + \beta_{1m}F_m + \epsilon_1 \\ x_2 - \mu_2 = \beta_{21}F_1 + \dots + \beta_{2m}F_m + \epsilon_2 \\ x_3 - \mu_3 = \beta_{31}F_1 + \dots + \beta_{3m}F_m + \epsilon_3 \\ \dots \\ \dots \\ \dots \\ x_p - \mu_p = \beta_{p1}F_1 + \dots + \beta_{pm}F_m + \epsilon_p \end{pmatrix}$$

Where:

- i μ_i is mean of variable i
- ii β_{ij} is $(p \times m)$ factor loading of the i^{th} variable on the j^{th} factor.
- iii ϵ_i is $(p \times 1)$ the error associated with i^{th} response X_i .
- iv F_j refers to the $m \times 1$ of the latent(unobservable) variables

The matrix notation

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix} - \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} = \begin{pmatrix} \beta_{11} & \cdot & \cdot & \cdot & \beta_{1m} \\ \beta_{21} & \cdot & \cdot & \cdot & \beta_{2m} \\ \vdots & & & & \\ \beta_{p1} & \cdot & \cdot & \cdot & \beta_{pm} \end{pmatrix} \times \begin{pmatrix} F_1 \\ F_2 \\ \vdots \\ F_m \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_p \end{pmatrix} \quad (2)$$

$$X - \mu = LF + \epsilon \quad (3)$$

Where:

μ is mean of variable i

L is the matrix of factor loadings

F refers to the matrix of the latent(unobservable) variables

ϵ is the matrix of the residuals

Assumptions of the the FA model

- i $E(F) = 0$
- ii $E(\epsilon) = 0$
- iii $Cov(\epsilon) = \Psi$ where Ψ is a diagonal matrix

$$\begin{pmatrix} \psi_1 & 0 & \cdots & 0 \\ 0 & \psi_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \psi_p \end{pmatrix}$$

- iv F and ϵ is independent.

0.1.4 Covariance Structure for the Orthogonal Factor Model

From equation (3) above, the orthogonal factor model implies a covariance structure for X.

$$\begin{aligned} (X - \mu)(X - \mu)' &= (LF + \epsilon)(LF + \epsilon)' \\ &= (LF + \epsilon)((LF)' + \epsilon') \\ &= LF(LF)' + \epsilon(LF)' + LF\epsilon' + \epsilon\epsilon' \end{aligned}$$

Therefore;

$$\begin{aligned}\Sigma &= Cov(X) = E(X - \mu)(X - \mu)' \\ &= LE(FF')L' + E(\epsilon F')L' + LE(F\epsilon') + E(\epsilon\epsilon') \\ &= LL' + \Psi\end{aligned}$$

similarly;

$$Cov(X, F) = L$$

The model $X - \mu = LF + \epsilon$ is linear in the common factors.

The portion of the variance of the i^{th} variable contributed by the common factors is called **the i^{th} communality**.

The portion of the variance due to the residual factor is referred to as **unique variance**.

As an illustration consider the equation below,

$$\underbrace{\sigma_{ii}}_{Var(X_i)} = \underbrace{\beta_{11}^2 + \beta_{12}^2 + \dots + \beta_{1m}^2}_{Communality} + \underbrace{\psi_i}_{Unique\ variance}$$

Thus,

$$Var(X_i) = Communality + Uniquevariance$$

0.2 Estimation of Loadings and Communalities

Given observations $x_1, x_2, \dots, x_n$ on p generally correlated variables, factor analysis seeks to answer the question, Does the factor model, with a small number of factors, adequately represent the data? In essence, we tackle this statistical model building problem by trying to verify the covariance relationship.

0.2.1 Principal components method

From a random sample $X_1, X_2, \dots, X_p$ we obtain the sample covariance matrix S and then we attempt to find an estimator $\hat{L}$ that will approximate the expression $\Sigma = LL' + \Psi$ with S instead of Σ i.e

$$S \approx \hat{L}L' + \Psi$$

In Principal factor method we ignore Ψ and factor S into $S = \hat{L}L'$; in order to factor S we use spectral decomposition

$$S = PDP'$$

where;

P is the normalized eigenvectors of S as columns

$$D = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_p \end{pmatrix}$$

L is estimated as

$$\hat{L} = [\sqrt{\lambda_1}e_1, \sqrt{\lambda_2}e_2, \dots, \sqrt{\lambda_k}e_k]$$

The i^{th} diagonal element of $\hat{L}L'$ is the sum of squares of the i^{th} row of $\hat{L}$.

The sums of squares of rows and columns of $\hat{L}$ are equal to the communalities and eigenvalues respectively i.e

$$\hat{h}_i^2 = \sum_{j=1}^k \hat{\beta}_{ij}^2$$

which is the sum of square of the i^{th} row of $\hat{L}$.

The sum of squares of the j^{th} column of $\hat{L}$ is the j^{th} eigenvalue of S i.e λ_j

The variance of the i^{th} variable is partitioned into a part due to the factors and due to uniquely to the variable

$$= \hat{h}_{i1}^2 + \hat{h}_{i2}^2 + \dots + \hat{h}_{ik}^2 + \Psi_i$$

Thus the j^{th} factor contributes $\hat{h}_{ij}^2$ to s_{ii} . The ontribution of the j^{th} factor to the total sample variance $tr(S) = s_{11} + s_{22} + \dots + s_{pp}$ is therefore;

$$= \sum_{j=1}^p \hat{\beta}_{ij}^2 = \hat{\beta}_{1j}^2 + \hat{\beta}_{2j}^2 + \dots + \hat{\beta}_{pj}^2$$

Which is the sum of squares of loadings in the coulumn of $\hat{L}$, which is is the same as the j^{th} eigen value.

The proportion of the total sample variance due to the j^{th} factor is

$$\frac{\lambda_j}{tr(S)}$$

and if the correlation matrix is used, the above proportion is;

$$\frac{\lambda_j}{p}$$

Note:

There is always some prior expectation concerning the loadings. In particular it is expected that some loadings will be close to zero while others will be negative or positive and substantially different from zero. For this reason FA usually proceeds in two stages.

1. one set of loadings β_{ij} is calculated which yields theoritical variances and covariances that fit the observed ones closely as possible according to certain criterion. These loading may however not agree with the prior expectation.
2. The first loading are rotated in an effort to arrive at another set that fit equally well the observed variances and covariances but are consistent with prior expectation or easily interpreted.

Illustration

As an illustration, let's consider the below example;

Patients seeking treatment from a certain psychiatry hospital are subjected to three tests on their trends in sleep, happiness and tiredness to tell the severity of their depression. Let Y1, Y2 and Y3 respectively represent the scores from the 3 tests respectively. The data consists of 5 patients (in a 10 point scale) as below,

Patient's Number	Y1	Y2	Y3
1	3	6	5
2	7	3	3
3	10	9	8
4	3	9	7
5	10	6	5

Table 1: Patient scores

It is suggested that the scores are functions of two underlying factors(symptoms) F1 and F2 i.e Cognitive symptoms and Affective symptom respectively. Each variable Y is assumed to linearly related to the two factors

$$Y1 = \mu_1 + \beta_{11}F1 + \beta_{12}F2 + \epsilon_1$$

$$Y2 = \mu_2 + \beta_{21}F1 + \beta_{22}F2 + \epsilon_2$$

$$Y3 = \mu_3 + \beta_{31}F1 + \beta_{32}F2 + \epsilon_3$$

In the measurement of severity of depression, trend in sleep point patients cognitive symptom while the level of happiness and tiredness feeling signify the Affective symptoms of depression. The visualization of the loading is represented as;

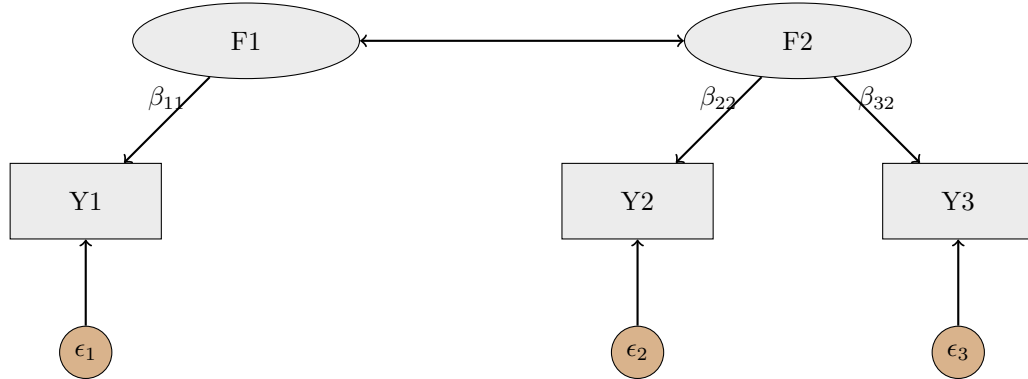


Figure 1: Visualization

$$\text{Thus } Var(Y_i) = \underbrace{\beta_{i1}^2 + \beta_{i2}^2}_{\text{Communality}} + \underbrace{\psi_i}_{\text{specific variance}}$$

If the factors were perfect predictors of the tests then $\epsilon_1 = \epsilon_2 = \epsilon_3 = 0$ and consequently, $\psi_1 = \psi_2 = \psi_3 = 0$

$$Cov(Y_i, Y_j) = \beta_{i1}\beta_{j1} + \beta_{i2}\beta_{j2}$$

Therefore our observed variance- covariance matrix S is;

	Y1	Y2	Y3
Y1	9.84	-0.36	0.44
Y2	-0.36	5.04	3.84
Y3	0.44	3.84	3.04

We determine the first set of loadings using principle component method that seeks values of the loadings that bring the the estimates of the total communalities as close as possible to the total observed variance(Covariance are ignored).

From our illustration, the elements of the principal components method are arranged as follows:

Variable	Observed Variance	Communality
Y1	S_1^2	$\beta_{11}^2 + \beta_{12}^2$
Y2	S_2^2	$\beta_{21}^2 + \beta_{22}^2$
Y3	S_3^2	$\beta_{31}^2 + \beta_{32}^2$
Total	T_0	T_t

The principal components method determines the values of the β_{ij} 's which makes the total communality T_t approximate as possible the sum of the observed varianciences, T_0 . Thus, from our illustration,

Variable	Observed Variance	Loadings on (F_1, β_{i1})	Loadings on (F_2, β_{i2})	Communality
Sleep Y1	9.84	3.136773	0.023799	9.98399
Happiness Y2	5.04	-0.312190	2.237859	5.0255
Tiredness Y3	3.04	0.127697	1.1731884	3.0157
Total	17.92	9.873125 *	8.007977 *	17.881

Table 2: Solution

From the solution in the table 2 above, we notice that the solution supports our expectation. The loadings on F_1 are relatively large for Y1 but close to zero for Y2 and Y3; loadings on F_2 are close to zero for Y1 but relatively high for Y2 and Y3. Implying F1 could be interperated as the cognitive symptom and F2 as Affective symptom.